Anjali Vathanakumaran

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Education

University of Waterloo, Candidate for BASc in Computer Engineering, Mechatronics Option, Class of 2028
Relevant Coursework: Embedded Microprocessor Systems, Digital Hardware Systems, Real-Time Operating Systems

Technical Skills

Languages: C, C++, SystemVerilog, VHDL, RISC-V Assembly, Python, HTML, CSS, Java, JavaScript
Protocols & Interfaces: UART, SPI, I2C, CAN, USB, PWM, TCP/Ethernet
Tools & Platforms: AMD Vivado, KV260, STM32, ESP32, Arduino, Altera FPGA, KiCad, Altium, Proteus, MATLAB, Git/GitHub

Work Experience

Firmware Team Lead | *Electrium Mobility, Waterloo, ON* December 2025 – September 2026

- Leading the development of a **BLDC** electronic speed controller (ESC), working on embedded motor control firmware involving state-machine-based control, PWM generation, sensor feedback, and closed-loop control validation.

Computer Vision Engineering Intern | *Martinrea International Inc, Vaughan, ON* May 2026 – August 2026

- Engineered a **React**/**HTML**-based front-end and **C++** backend to bring the standardized **Python**-based vision system into a live inspection application, integrating a **Django-based** data-offloading pipeline to sync edge-device results to a centralized server.
- Designed a human-in-the-loop vision system accommodating real-time inspection for **13 assemblies** to reduce manual inspection time, selecting cameras and hardware while evaluating and integrating **YOLO**, **TensorRT**, and **ONNX** models.

Computer Vision Application Intern | *Martinrea International Inc, Vaughan, ON* September 2025 – December 2025

- Re-architected and standardized a production-grade **Python** codebase for industrial **ML-based computer vision systems** into a **multiprocessing** architecture, adopted as the core codebase for CV deployments across manufacturing plants globally.
- Adapted the vision system for a **conveyor belt production line**, developing a **ghost-box tracking algorithm** to maintain part identity under variable camera frame rates, preventing duplicate or dropped detections that could stall production.
- Developed and trained a **three-stage model pipeline** (**detection**, **classification**, and **segmentation**) using machine learning architectures to perform sequential quality checks on each part, reducing defect detection false negatives to **0.77%**.

Software Developer Intern | *The Honest Consultants, Toronto, ON* January 2025 – April 2025

- Designed and implemented scalable backend services using **Node.js**, building dynamic data synchronization pipelines between cloud services, implementing schema-flexible **GraphQL** queries to support various client data models.
- Developed internal tooling dashboards using **HTML/CSS** and server-rendered templates, automating client configuration workflows and reducing manual operational overhead.

Software Engineering Intern | *Graze AI, Toronto, ON* May 2024 – August 2024

- Built and scaled production-grade web scraping automation systems in **Python**, engineering robust data pipelines that extracted millions of data points and reduced data sourcing costs by over **$10k** annually.
- Architected and iteratively benchmarked multiple scraping pipelines using **Django, Selenium and Scrapy**, selecting optimal approaches to reduce runtime by **60%** and improve reliability through automated data enrichment.

Projects

Real-Time Executive (RTX) Kernel January 2026

- Built a **multitasking kernel** in **C** on the **STM32** (ARM Cortex-M4), implementing task scheduling APIs, TCB-based state management, and interrupt-driven context switching to support safe concurrent execution.

Four-Letter Wordle January 2026

- Implemented a custom Wordle game in **SystemVerilog** on the **KV260**, designing finite state machines and synthesizable control logic to manage user input, state transitions, and display output on FPGA hardware.

Water Reservoir Monitoring System May 2025 – August 2025

- Designed an embedded irrigation control prototype on an **STM32**, using a potentiometer to set **PWM** duty cycle for DC motor-driven watering output, a servo to route flow to the selected field, and distance sensors for water-level sensing.
- Built a custom timer board with dual **7-segment displays** and switches to set and count down each field's watering duration, verifying signal timing with an **oscilloscope**; translated the prototype into a custom PCB in **Proteus**.